# WaveMamba: Wavelet-Driven Mamba Fusion for RGB-Infrared Object Detection

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## Abstract

*Leveraging the complementary characteristics of visible (RGB) and infrared (IR) imagery offers significant potential for improving object detection. In this paper, we propose WaveMamba, a cross-modality fusion method that efficiently integrates the unique and complementary frequency features of RGB and IR decomposed by Discrete Wavelet Transform (DWT). An improved detection head incorporating the Inverse Discrete Wavelet Transform (IDWT) is also proposed to reduce information loss and produce the final detection results. The core of our approach is the introduction of WaveMamba Fusion Block (WMFB), which facilitates comprehensive fusion across low-/high-frequency sub-bands. Within WMFB, the Low-frequency Mamba Fusion Block (LMFB), built upon the Mamba framework, first performs initial low-frequency feature fusion with channel swapping, followed by deep fusion with an advanced gated attention mechanism for enhanced integration. High-frequency features are enhanced using a strategy that applies an "absolute maximum" fusion approach. These advancements lead to significant performance gains, with our method surpassing state-of-the-art approaches and achieving average mAP improvements of 4.5% on four benchmarks.*

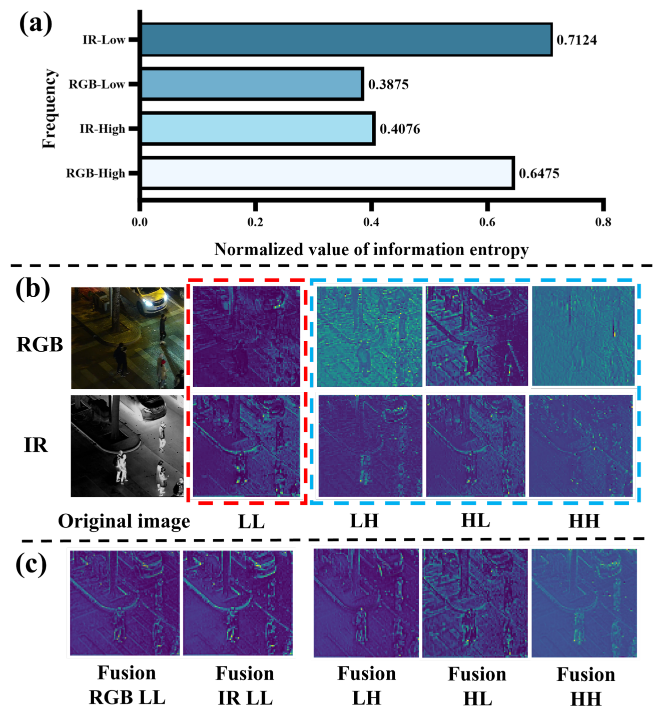


Figure 1. (a) shows the averaged normalized information entropy of low- and high-frequency components from RGB and IR images in $M^3$FD val dataset using DWT. (b) depicts the RGB and IR features filtered by DWT in two directions, with low-frequency (L) and high-frequency (H) components obtained in each direction, resulting in four feature sub-bands: LL, LH, HL, and HH. Red and blue dashed boxes highlight the low-frequency and high-frequency components, respectively. The first row shows RGB and the second row shows IR components. (c) shows the result of fusion of each frequency.

## 1. Introduction

The fusion of multi-spectral features from visible (RGB) and infrared (IR) imagery has emerged as a pivotal technique in computer vision. This is especially applicable to scenarios in challenging environments where traditional methods alone may be insufficient [21, 56, 63]. RGB-based methods often encounter limitations in low-light or adverse

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weather conditions, as RGB images tend to lose detail and contrast. Conversely, IR images excel in these conditions by capturing thermal signatures that are independent of visible light, thus offering complementary information [15, 36, 76]. However, the integration of these distinct modalities (*i.e.*, RGB and IR) into a cohesive framework poses significant challenges, including the effective fusion of heterogeneous data and maintaining computational efficiency.

Over the past decades, numerous approaches have been developed to integrate information from visible and infrared images for cross-modality object detection. These approaches can be typically classified into CNN-based [31, 49, 57, 62], Transformer-based [7, 23, 39, 72], and Mamba-based [10, 34, 84] methods. While CNN-based methods have demonstrated success, they often struggle to capture long-range dependencies, leading to a growing interest in Transformer-based techniques. Recently, the high computational complexity of Transformer models has further motivated the development of Mamba-based methods, offering a more efficient alternative. For instance, Gao et al. [10] introduced MambaST, an efficient cross-spectral spatial-temporal fusion pipeline that enhances pedestrian detection by employing a multi-head hierarchical patching and aggregation structure to fuse visible and thermal imagery. Therefore, we utilize the Mamba framework [33] as the foundation for our feature fusion due to its ability to balance efficiency and performance.

While existing cross-modality object detection methods have shown promising performance, research on effectively using inherent features between modalities for complementary advantages to enhance performance remains insufficient. We endeavor to explore answers from the frequency domain. Specifically, as shown in Fig. 1(a), applying Discrete Wavelet Transform (DWT) to $M^3$FD validation set reveals that IR images exhibit higher normalized information entropy in the low-frequency sub-band, while RGB images exhibit greater entropy in the high-frequency subbands. The value of information entropy reflects the amount of information carried by the image. Generally speaking, the higher the information entropy, the richer the content of image is [58]. This statistical observation suggests that RGB images are more adept at capturing high-frequency details, in contrast to IR images, which predominantly encapsulate low-frequency signals. Furthermore, Fig. 1(b) illustrates the DWT outputs of P2-level RGB and IR features derived from the YOLOv8 backbone trained on each respective modality. The low- and high-frequency components are delineated by the red and blue dashed box, respectively. In the low-frequency region, IR features manifest with clearer shapes, while RGB features present more pronounced edges and contours in the high-frequency region, affirming their distinct frequency characteristics—an observation that aligns with prior studies [59]. Building on these insights, *we aim to exploit the unique and complementary frequency characteristics of IR and RGB modalities to achieve enhanced feature fusion, thereby improving object detection performance.*

In this paper, we propose WaveMamba, as illustrated in Fig. 2, which leverages the DWT to map features into the wavelet domain, exploiting the complementary frequency characteristics of RGB and IR for efficient feature fusion. An improved detection head with Inverse DWT (IDWT) is then used to generate the final detection results, leading to improved detection accuracy. The main innovation of WaveMamba lies in WaveMamba Fusion Block (WMFB), which comprises two key components: the Low-frequency Mamba Fusion Block (LMFB) and the High-Frequency Enhancement (HFE) strategy. LMFB utilizes the exceptional low-frequency modeling capability of Mamba [35] to integrate low-frequency information. It comprises two modules: the Shallow Fusion Mamba (SFM) module, which facilitates lightweight cross-modality interaction by swapping portions of RGB and IR channels, and the Deep Fusion Mamba (DFM) module, which further refines the fusion process using a gated attention mechanism to filter out redundant information and improve the quality of the low-frequency feature fusion. Meanwhile, the HFE strategy enhances the selection and retention of critical details within high-frequency features by adopting an "absolute maximum" fusion approach. This combination of innovations significantly improves the fusion of features from both modalities, as shown in Fig. 1(c). The low-frequency features of RGB fusion are enriched, while the IR fusion features effectively suppress redundant noise in the originally low-frequency characteristics of IR, such as road shadows. Meanwhile, the fused high-frequency features become more detailed and pronounced. In summary, this paper's main contributions are:

- We propose WaveMamba, which leverages the intrinsic advantageous frequency characteristics of different modalities to fuse features, facilitating sufficient feature interaction and enhancing the performance of RGB-IR object detection.
- WaveMamba Fusion Block efficiently integrates distinct frequency information, including a Low-frequency Mamba Fusion Block that combines low-frequency features and a High-Frequency Enhancement strategy to capture details without adding computational complexity. An improved YOLOv8 detection head utilizes IDWT for final detection results.
- Extensive experiments on four public datasets demonstrate that our approach achieves state-of-the-art performance in cross-modality object detection tasks, with an average $m$AP exceeding that of the second-best method by $4.5\%$.

## 2. Related Works

**Multi-modality Object Detection.** Various multi-modality object detection methods have been proposed based on the complementary information from RGB-IR images. So far, these methods can be categorized into two main directions: pixel-based and feature-based. Pixel-based methods such as CDDFuse [82] and SuperFusion [45], involve fusing two images at the pixel level to obtain a single fused image, upon which object detection is then performed. Feature-based methods fuse the output from a certain stage of the two-stream detector, such as the early or middle stage features extracted by the backbone [2, 27, 29, 48, 57] or the results of the detector (*i.e.*, late fusion) [5, 25, 28, 30]. However, the fusion modules designed in these methods are predominantly based on CNN or Transformer architectures. CNN's are constrained by their local receptive fields, while Transformer's face inefficiencies in model detection due to their complexity. Recently, the DMM framework, as described in [84], incorporates fusion modules designed to optimize and merge features from two modalities using Mamba, leading to successful outcomes in oriented object detection. Despite the excellent performance achieved by these methods, further enhancement in detection performance can be realized by utilizing distinct characteristics in the frequency domain between modalities.

**Wavelet Transform in Computer Vision.** As wavelet transform can achieve lossless up- and down-sampling and obtain frequency domain information of images, it has been widely applied to various computer vision tasks [14, 47, 70, 86], resulting in performance improvements. WTConv [9] achieves a significant increase in the convolutional receptive field by integrating wavelet and inverse wavelet transforms into the convolution process. WF-Diff [77] attains cutting-edge performance in underwater image restoration by using wavelet transform to obtain the high- and low-frequency components of the image after preliminarily enhanced, and then sending them into two separate diffusion branches to adjust the high- and low-frequency information. DCFNet [60] realizes an improvement in the quality of fused images for the task of RGB and infrared image fusion. It incorporates wavelet transform and inverse wavelet transform as pre-processing steps for the Encoder and Decoder, respectively. However, these methods fail to distinguish different frequency domain characteristics of low- and high-frequency components, adopting either identical processing methods for both or without processing the high-frequency components altogether. Consequently, the extraction of frequency domain information is inadequate.

Unlike the aforementioned methods, we utilize wavelet transform and its inverse transform in cross-modality object detection, designing distinct processing modules for high- and low-frequency components, resulting in significant improvements in detection efficiency and performance.

## 3. Methods

### 3.1. Preliminaries

**Discrete Wavelet Transform.** Haar wavelet transform is a standard DWT due to its simplicity and efficiency, and we set it as the default. For an image $I$ , the one-level transform applies depth-wise convolution using low-pass $L = \frac{1}{\sqrt{2}}[1, 1]$ and high-pass $H = \frac{1}{\sqrt{2}}[1, -1]$ filters, followed by a down-sampling step that reduces the resolution by a factor of 2. Extending to two dimensions involves four filters, and we have sub-bands for $I$ as below:

$$F_L, F_H = \mathrm{DWT}(I), \tag{1}$$

where $\mathrm{DWT}(\cdot)$ denotes the Haar wavelet transform, sub-bands $F_L = \{F_{LL}\}$ captures low-frequency information and $F_H = \{F_{LH}, F_{HL}, F_{HH}\}$ captures high-frequency details and noise.

**2D Selective Scan.** In linear time-invariant systems, state space models (SSMs) process 1D input signals through state variables and genetate outputs using ODEs. Mamba [11] enhances this modeling approach, but its direct application to vision tasks is limited by the mismatch between 2D visual data and 1D language sequences. To address this, the 2D Selective Scan (SS2D) mechanism introduced in [33] expands image patches in four directions, creating independent sequences. Each feature sequence is processed using the Selective Scan Space State Sequential Model (S6) [11], and the sequences are then aggregated to reconstruct the 2D feature map.

### 3.2. WaveMamba

The architecture of our model, as depicted in Fig. 2, consists of a dual-stream feature extraction backbone augmented with the DWT, three WaveMamba Fusion Blocks, and an improved YOLOv8 detection head that incorporates the IDWT.

Given the input RGB images $I_{\mathrm{RGB}}$ and infrared images $I_{\mathrm{IR}}$, our model utilizes the first two layers of the dual-stream backbone to extract local features $f^i_{\mathrm{RGB}}$ and $f^i_{\mathrm{IR}}$ where the superscript $i \in \{1, 2\}$ indicates the layer. $f^2_{\mathrm{RGB}}$ and $f^2_{\mathrm{IR}}$ undergo the Haar wavelet transformation and result in corresponding features in the wavelet domain:

$$\begin{aligned} F^2_{L,\mathrm{RGB}}, F^2_{H,\mathrm{RGB}} &= \mathrm{DWT}(f^2_{\mathrm{RGB}}), \\ F^2_{L,\mathrm{IR}}, F^2_{H,\mathrm{IR}} &= \mathrm{DWT}(f^2_{\mathrm{IR}}), \end{aligned} \tag{2}$$

where $F^2_{L,\mathrm{RGB}}$ and $F^2_{L,\mathrm{IR}}$ denote the low-frequency sub-bands, while $F^2_{H,\mathrm{RGB}}$ and $F^2_{H,\mathrm{IR}}$ represent the high-frequency sub-bands of the RGB and infrared modalities, respectively. With these frequency features combined with our proposed WMFB, we obtain fused features $F^3_{\mathrm{RGB}}$ and $F^3_{\mathrm{IR}}$. Like [9], multi-level wavelet transformations are used

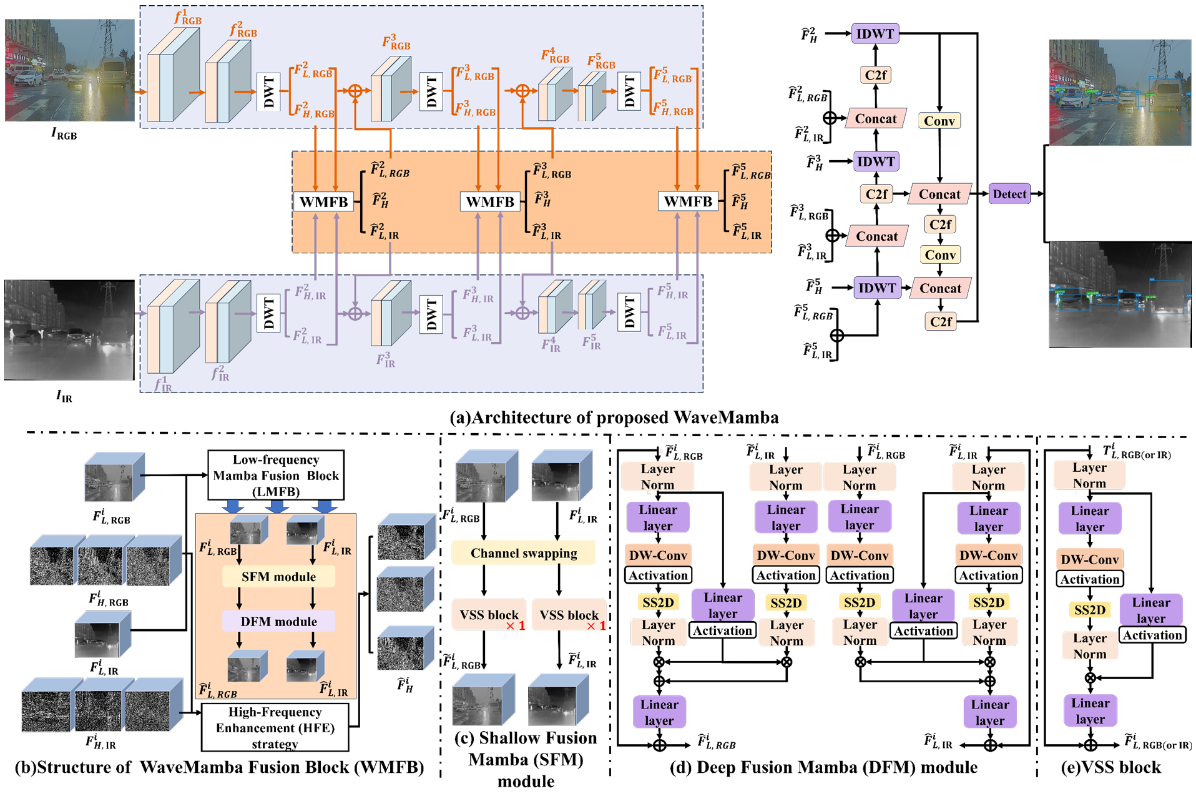


Figure 2. Illustrative diagram of the complete network and sub-modules. (a) shows the overall architecture of proposed WaveMamba network, which mainly includes the dual-stream feature extraction backbone improved by adding DWT, three WaveMamba Fusion Blocks and the improved YOLOv8 head augmented with IDWT. (b) shows the detailed structure of the WMFB, which includes the LMFB and the HFE strategy. (c), (d) and (e) show the detailed structure of the SFM module, DFM module and VSS block, respectively.

in our framework to achieve more detailed multi-scale features. Specifically, DWT and WMFB are applied to the fused features at the $3^{rd}$ and $5^{th}$ layers. Finally, the fused low- and high-frequency sub-bands (*i.e.*, all the output of WMFB) are fed into the improved YOLOv8 detection head, where IDWT is used for feature up-sampling, resulting in the final detection outputs. In the following sections, we introduce WMFB in subsec. 3.3 and the improved YOLOv8 head in subsec. 3.4.

### 3.3. WaveMamba Fusion Block

As shown in Fig. 1, we find that RGB images typically possess more high-frequency details, while IR images primarily capture low-frequency information. To effectively leverage the complementary frequency characteristics of both RGB and IR for feature fusion, we develop the WMFB. Specifically, WMFB consists of the Low-frequency Mamba Fusion Block and the High-Frequency Enhancement strategy for low- and high-frequency fusion, respectively.

#### 3.3.1. Low-frequency Mamba Fusion Block

LMFB processes low-frequency features $F^i_{L,\text{RGB}}$, $F^i_{L,\text{IR}}$ in $i^{th}$ layer to integrate low-frequency information from shallow to deep. In this case, LMFB designs the shallow fusion mamba module and deep fusion mamba module.

**SFM Module.** This module is designed to enhance the interaction of low-frequency features across modalities for shallow feature fusion. It utilizes a channel swapping operation [13] in conjunction with VSS blocks [33]. We first integrate information from distinct channels to construct cross-modality feature correlations, thereby enriching the diversity of channel features and improving fusion performance. We begin with the channel swapping to derive new local features for low-frequency RGB and IR modalities, respectively. This can be formulated as follows:

$$T^i_{L,\text{RGB}}, T^i_{L,\text{IR}} = \text{CS}(F^i_{L,\text{RGB}}, F^i_{L,\text{IR}}), \tag{3}$$

where $\text{CS}(\cdot)$ denotes the channel swapping operation. This operation is implemented through channel splitting and concatenation to enhance the diversity of channel features. To further enhance the fused features, two VSS blocks are applied to $T^i_{L,\text{RGB}}$ and $T^i_{L,\text{IR}}$ separately as below:

$$\tilde{F}^i_{L,\text{RGB}} = \text{VSS}(T^i_{L,\text{RGB}}), \quad \tilde{F}^i_{L,\text{IR}} = \text{VSS}(T^i_{L,\text{IR}}), \tag{4}$$

where $\text{VSS}(\cdot)$ refers to the VSS block, as shown in Fig. 2(e). The outputs $\tilde{F}^i_{L,\text{RGB}}$ and $\tilde{F}^i_{L,\text{IR}}$ represent the shallow fused features from the RGB and IR modalities, respectively.

**DFM Module.** To further facilitate the interaction and deep fusion of low-frequency features between modalities, we introduce DFM based on the output of SFM as shown in Fig. 2(d). In DFM, the SFM outputs $(\tilde{F}^i_{L,\mathrm{RGB}}, \tilde{F}^i_{L,\mathrm{IR}})$ alternate between primary and auxiliary roles. The primary modality is linearly embedded and split into two streams: one passes through a 3×3 depth-wise convolution, SiLU activation, the core SS2D module, and layer normalization; the other undergoes only SiLU activation. The auxiliary modality follows the primary stream's first pathway. Given the complexity and diversity of features present across different modalities, it is crucial to selectively emphasize unique and complementary characteristics while minimizing redundancy. On this basis, we introduce a gated attention mechanism to effectively facilitate the learning of complementary features while suppressing redundant features. Specifically, this mechanism uses the output of the second information flow of the primary modality to regulate the outputs of both the first information flow of the primary modality and the auxiliary modality. By summing these regulated results, we obtain the final output of the DFM module for $i^{th}$ layer, formalized as follows:

$$\hat{F}^i_{L,\mathrm{RGB}}, \hat{F}^i_{L,\mathrm{IR}} = \mathrm{DFM}(\tilde{F}^i_{L,\mathrm{RGB}}, \tilde{F}^i_{L,\mathrm{IR}}). \tag{5}$$

#### 3.3.2. High-Frequency Enhancement Strategy

High-frequency sub-bands capture fine details in the original features, where larger absolute values indicate more significant information. To select and retain critical details, we employ a HFE strategy, which efficiently integrates high-frequency information by choosing pixels with larger absolute values. The HFE strategy for $i^{th}$ layer high-frequency sub-bands is defined as follows:

$$\begin{aligned} F^i_H &= \mathrm{Mask}(|F^i_{H,\mathrm{RGB}}| - |F^i_{H,\mathrm{IR}}|) \odot F^i_{H,\mathrm{RGB}} \\ &+ \mathrm{Mask}(|F^i_{H,\mathrm{IR}}| - |F^i_{H,\mathrm{RGB}}|) \odot F^i_{H,\mathrm{IR}}, \end{aligned} \tag{6}$$

where $\odot$ is element-wise product operation. Mask$(\cdot)$ generates an element-wise binary mask, where element is set to 1 if its corresponding input is greater than 0; otherwise, it is set to 0. Note that $F^i_H = \{F^i_{LH}, F^i_{HL}, F^i_{HH}\}$ based on Eq. 1. In this case, we process $F^i_{LH}, F^i_{HL}, F^i_{HH}$ independently and construct $F^i_H$. Lastly, $F^i_H$ is the resulting merged high-frequency subbands.

### 3.4. Improved YOLOv8 head

Since our network architecture utilizes multilevel wavelet transformations to obtain multi-scale features, we intuitively design a corresponding YOLOv8 detection head structure using IDWT. Compared to traditional upsampling, IDWT reduces information loss and significantly improves the accuracy of detection results.

As shown in Fig. 2(a), initially, we aggregate the fused low-frequency features from two modalities to get the final low-frequency feature. Subsequently, IDWT is utilized on this low-frequency feature alongside the fused high-frequency features. Aside from this modification, the rest in the detection head remain consistent with those in the original YOLOv8. Through these operations, we derive the ultimate detection results.

## 4. Experiments

### 4.1. Experimental Setups

We evaluate our model on six datasets with $mAP_{50}$ and $mAP$: $M^3$FD [32], DroneVehicle [44], LLVIP [17], FLIR-Aligned [75] in the main paper and VEIDA [37], KAIST [16] datasets in the supplementary.

Moreover, for the FLIR-Aligned dataset, we also employ precision, recall, and F1 score as evaluation metrics.

Our model and training framework are based on a modified dual-stream framework of the YOLO architecture by Ultralytics [50]. To verify the universality of our method, we implement our approach with three standard backbones: **ResNet50**, **YOLOv5** and **YOLOv8**, following existing methods [39]. The loss function and training pipeline are identical to those used in standard YOLOv8 training [50]. More details can be found in the supplementary materials.

### 4.2. Comparison with SOTA Results

**M³FD Dataset.** $M^3$FD is a multi-class object detection dataset in extreme weather conditions. The results on $M^3$FD are summarized in Table 1. Compared to existing state-of-the-art methods, our YOLOv8-based method achieves the highest performance, with improvements of $5.5\%$ and $5.1\%$ in $mAP_{50}$ and $mAP$, respectively. Similarly, our method using ResNet50 and YOLOv5 as backbones also outperforms other methods using the same backbones. For performance in different categories, our method with different backbones consistently achieves top-three rankings in most cases. Specifically, our method outperforms the fourth-ranked method by $1.1\%$, $6.0\%$, and $8.5\%$ in the categories of cars, lamps, and people, respectively. Note that even without special design for rare classes, our method still achieves similar performance in these rare classes (e.g., "Bus", "Truck", "Motorcycle"). Compared to the single-modal YOLOv8l model, our fusion models achieve significant performance improvements, demonstrating the effectiveness of our method in integrating beneficial information from both the IR and RGB modalities. As our method effectively leverages the global structural information from the low-frequency sub-band of IR and the detailed information, such as edge textures of objects, from the high-frequency sub-bands of RGB, we demonstrate exceptional detection performance under extreme weather conditions. Furthermore, the consistent improvement across the three backbones indicates that our fusion approach is versatile and adaptable to various feature

| Methods | Backbone | $mAP_{50}$ | $mAP$ | Car | Bus | Lamp | People | Truck | Motorcycle |
|---|---|---|---|---|---|---|---|---|---|
| (LOPET'24) LG+MDA [69] | ResNet50 | 66.4 | 40.2 | 83.0 | 84.4 | 37.7 | 62.5 | 75.5 | 54.6 |
| (ECCV'24) DAMSDet [12] | ResNet50 | 80.2 | 52.9 | - | - | - | - | - | - |
| Ours | ResNet50 | 90.9 | 62.3 | 95.6 | 96.4 | 92.8 | 91.6 | 84.8 | 84.4 |
| (CVPR'22) RFNet [65] | YOLOv5 | 79.4 | 53.2 | 91.1 | 78.2 | 85.0 | 79.4 | 69.9 | 72.8 |
| (CVPR'22) TarDAL [32] | YOLOv5 | 80.5 | 54.1 | 94.8 | 81.3 | 87.1 | 81.5 | 68.7 | 69.3 |
| (IJCAI'20) DIDFuse [80] | YOLOv5 | 77.2 | 51.4 | 91.0 | 78.3 | 83.2 | 78.5 | 66.3 | 66.3 |
| (IJCV'21) SDNet [74] | YOLOv5 | 79.3 | 52.9 | 91.1 | 78.2 | 85.0 | 79.4 | 69.0 | 72.8 |
| (MM'22) DetFusion [43] | YOLOv5 | 80.8 | 53.8 | 92.5 | 83.0 | 87.8 | 80.8 | 71.4 | 69.4 |
| (CVPR'23) CDDFuse [82] | YOLOv5 | 80.5 | 53.6 | 91.6 | 82.4 | 86.0 | 81.1 | 71.2 | 71.1 |
| (MM'23) IGNet [26] | YOLOv5 | 81.5 | 54.5 | 92.8 | 82.4 | 86.9 | 81.6 | 72.1 | 73.0 |
| (TCSVT'24) MMFN [67] | YOLOv5 | 86.2 | 57.4 | 93.2 | 92.1 | 87.6 | 83.0 | 87.4 | 73.7 |
| (CVPR'24) EMMA [83] | YOLOv5 | 82.9 | 55.4 | 93.5 | 83.2 | 87.7 | 82.0 | 73.5 | 77.7 |
| (TIM'24) KCDNet [51] | YOLOv5 | 83.2 | 56.3 | 90.9 | 88.4 | 80.3 | 83.3 | 72.1 | 84.1 |
| (TJRS'24) CRSIOD [53] | YOLOv5 | 84.0 | 57.2 | 92.2 | 93.3 | 80.5 | 78.4 | 85.5 | 73.9 |
| Ours | YOLOv5 | 91.5 | 63.8 | 95.4 | 95.4 | 93.2 | 91.8 | 87.8 | 85.7 |
| YOLOv8l-IR [50] | YOLOv8 | 79.5 | 53.1 | 90.0 | 90.9 | 63.0 | 82.9 | 85.9 | 64.6 |
| YOLOv8l-RGB [50] | YOLOv8 | 80.9 | 52.5 | 91.2 | 92.9 | 75.3 | 70.6 | 86.0 | 69.6 |
| (JAS'22) SuperFusion [45] | YOLOv7 | 83.5 | 56.0 | 91.0 | 93.2 | 70.0 | 83.7 | 85.8 | 77.4 |
| (2024) TSJNet [19] | YOLOv7 | 86.0 | 58.9 | 91.8 | 95.7 | 70.4 | 81.8 | 89.3 | 86.8 |
| (YAC'24) RI-YOLO [4] | YOLOv8 | 83.6 | 56.6 | 91.6 | 88.5 | 75.7 | 84.1 | 87.8 | 73.5 |
| (Sensors'24) MRD-YOLO [40] | YOLOv8 | 86.6 | 59.3 | - | - | - | - | - | - |
| Ours | YOLOv8 | 92.1 | 64.4 | 95.9 | 95.6 | 93.8 | 92.6 | 88.6 | 86.1 |

Table 1. Comparison results with SOTA methods on $M^3$FD dataset. The best results are highlighted in red. The second and third best results are highlighted in green and blue, respectively.

| Methods | Backbone | $mAP_{50}$ | $mAP$ | Car | Bus | Truck | Freight-car | Van |
|---|---|---|---|---|---|---|---|---|
| (ECCV'22) TSFADet [73] | ResNet50 | 73.1 | 44.1 | 89.9 | 89.8 | 67.9 | 63.7 | 54.0 |
| (TGRS'24) $C^2$Former-$S^2$ANet [72] | ResNet50 | 74.2 | 47.3 | 90.2 | 89.8 | 68.3 | 64.4 | 58.5 |
| (GRSL'24) GLFNet [20] | ResNet50 | 71.4 | 54.8 | 90.3 | 88.0 | 72.7 | 53.6 | 52.6 |
| (JST'24) FFODNet [55] | ResNet50 | 76.3 | 56.9 | - | - | - | - | - |
| (2024) DMM [84] | ResNet50 | 77.2 | 55.8 | 90.4 | 88.7 | 77.8 | 63.0 | 66.0 |
| (TGRS'24) LF-MDet [41] | ResNet50 | 71.8 | 51.3 | - | - | - | - | - |
| Ours | ResNet50 | 79.3 | 59.9 | 94.6 | 90.2 | 79.6 | 68.0 | 64.1 |
| (YAC'24) CAFN-IA [66] | YOLOv5 | 69.3 | 56.1 | 89.1 | 90.8 | 62.0 | 57.3 | 47.1 |
| (MTA'23) SLBAF-Net [6] | YOLOv5 | 76.8 | 49.5 | 90.2 | 89.9 | 76.0 | 68.2 | 59.9 |
| (CVPR'24) CSOM-ODAF [3] | YOLOv5 | 77.1 | 50.1 | 90.1 | 89.8 | 75.6 | 68.2 | 61.8 |
| (RSL'24) multimodal DINO [42] | YOLOv5 | 72.5 | 50.3 | 89.5 | 88.8 | 75.4 | 54.3 | 54.3 |
| Ours | YOLOv5 | 79.5 | 60.2 | 94.8 | 90.1 | 80.1 | 68.2 | 64.3 |
| YOLOv8l-IR [50] | YOLOv8 | 71.9 | 49.1 | 93.4 | 91.9 | 69.3 | 53.7 | 51.1 |
| YOLOv8l-RGB [50] | YOLOv8 | 70.2 | 48.6 | 92.5 | 91.7 | 68.8 | 47.8 | 50.2 |
| (Sensors'23) Dual-YOLO [1] | YOLOv7 | 71.9 | 55.2 | 95.9 | 91.6 | 69.7 | 55.9 | 46.6 |
| (AEORS'24) YOLOFIV [52] | YOLOv8 | 64.7 | 53.1 | 95.9 | 91.6 | 64.2 | 34.6 | 37.3 |
| (Sensors'24) IV-YOLO [49] | YOLOv8 | 74.6 | 56.8 | 97.2 | 94.3 | 65.4 | 63.1 | 53.0 |
| (RS'24) DAAB-FFPN [54] | YOLOv8 | 75.2 | 56.3 | - | - | - | - | - |
| (TGRS'24) CRSIOD [53] | YOLOv8 | 73.2 | 50.8 | 95.6 | 92.2 | 71.7 | 50.5 | 55.8 |
| (IJAEOG'24) CMA [18] | YOLOv8 | 76.8 | 50.4 | 95.8 | 93.1 | 75.9 | 59.8 | 59.4 |
| Ours | YOLOv8 | 79.8 | 60.5 | 95.0 | 90.6 | 80.4 | 68.5 | 64.5 |

Table 2. Comparison results with SOTA methods on DroneVehicle dataset. The best results are highlighted in red. The second and third best results are highlighted in green and blue, respectively.

extraction methods, thus providing stable performance improvements.

**DroneVehicle Dataset.** DroneVehicle is a challenging remote sensing dataset with densely annotated images and small targets (e.g., "car", "bus", "van"). As shown in Table 2, our method achieves top-three rankings in both $mAP_{50}$ and $mAP$ metrics, surpassing the fourth-place method by 2.6% and 3.6%, respectively. Unlike other methods, which leverages domain-specific designs for remote sensing, our approach remains general yet still secures top-three positions in the categories of "truck" and "freight-car". Thanks to our unique low- and high-frequency fusion, our method excels at capturing edge details and achieves superior overall performance, highlighting its ability to enhance object detection across diverse scenarios without specialized remote sensing design.

**LLVIP Dataset.** LLVIP is a low-light dataset for pedestrian detection. As presented in Table 3, we compare the performance of our approach on LLVIP dataset against state-of-the-art single-modal and fusion models. Our method achieves superior performance when using the same backbone. In particular, our YOLOv8-based method achieves remarkable state-of-the-art results, with 98.3% in $mAP_{50}$ and 66.0% in $mAP$.

| Methods | Backbone | $mAP_{50}$ | $mAP$ |
|---|---|---|---|
| (CVPRW'23) CSAA [2] | ResNet50 | 94.3 | 54.2 |
| (2024) RSDet [79] | ResNet50 | 95.8 | 61.3 |
| (TIV'24) MMSANet [24] | ResNet50 | 96.4 | 62.5 |
| (CVPR'24) Text-IF [71] | Transformer | 94.1 | 60.2 |
| Ours | ResNet50 | 97.3 | 65.0 |
| (IF'23) DIVFusion [46] | YOLOv5 | 89.8 | 52.0 |
| (PR'24) ICAFusion [39] | YOLOv5 | 95.2 | 60.1 |
| Ours | YOLOv5 | 97.6 | 65.2 |
| YOLOv8l-IR [50] | YOLOv8 | 95.2 | 62.1 |
| YOLOv8l-RGB [50] | YOLOv8 | 91.9 | 54.0 |
| (ICHMS'24) MDMDet [68] | YOLOv7 | 96.5 | 61.5 |
| (RS'24) ACDF-YOLO [8] | YOLOv8 | 96.5 | 61.3 |
| (ICHMS'24) MSFF [68] | YOLOv8 | 96.8 | 61.9 |
| (Sensors'24) FAWDet [78] | YOLOv8 | 97.1 | 62.1 |
| Ours | YOLOv8 | 98.3 | 66.0 |

Table 3. Comparison results with SOTA methods on LLVIP dataset. The best results are highlighted in red. The second and third best results are highlighted in green and blue, respectively.

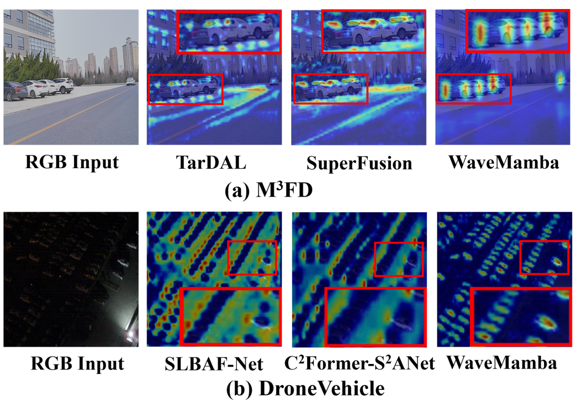


Figure 3. Heatmap visualization of several cross-modality object detection methods on $M^3$FD and DroneVehicle.

**FLIR-Aligned Dataset.** To demonstrate the advantages of our approach, we compare our methods with state-of-the-art methods with more metrics. As shown in Table 4, our method outperforms other fusion methods using the same backbone, achieving superior accuracy, while also reducing the size of the parameters and enhancing the speed of inference. For example, compared to CrossFormer, our method surpasses $5.8\%$ in precision, $6.9\%$ in recall and $6.3\%$ in the F1 score, and reduces the model size by $294.4$ million parameters and decreases the inference time by $35.9$ milliseconds. MFPT and ESSFN exhibit a similar trend. This benefit is due to our fusion design and the linear time complexity of Mamba. This further highlights the effectiveness of our approach in optimizing computational resources and improving detection performance.

### 4.3. Visualization

**Heatmaps.** We visualize the heatmap of our model's first inverse wavelet transform layer based on Grad-CAM [38] and compare it with other state-of-the-art methods [6, 32, 45, 72]. As shown in Fig. 3, our model effectively prioritizes the target by utilizing wavelet transform and feature fusion, minimizing unnecessary attention on the background. In contrast, others tend to focus on larger areas, including backgrounds, which renders small targets indistinguishable.

**Detection Results.** We visualize our detection results and compare them with other state-of-the-art methods [6, 32, 45, 72]. As shown in Fig. 4, our method reduces the number of missed or false detections in challenging scenarios (*i.e.*, adverse weather, low lighting, heavy occlusion, or dense detection conditions), achieving the superior detection performance.

### 4.4. Ablation Study

To verify the efficacy of each module and different fusion strategies, we perform ablation studies using the YOLOv8 backbone on the $M^3$FD dataset. More ablation studies such as the position and number of our WMFB are provided in the supplementary materials.

**Effects of the Improved Head.** We compare our improved head with the original YOLOv8 head in Table 5. As we can see, utilizing our improved head results in a $1.1\%$ improvement in $mAP_{50}$ and $mAP$, coupled with a decrease of 7.6M in the number of parameters. The achievement can be attributed to the use of DWT and IDWT rather than down- and up-sampling. The use of DWT and IDWT preserves more detailed information, reduces information loss, and enhances the capture and reconstruction of features at multiple scales, resulting in improved overall model performance [9, 61, 70].

**Effects of the SFM and DFM Modules.** In Table 6, we ablate the impact of SFM and DFM Modules by removing these two modules separately. Removing SFM leads to a decrease in performance by $1.5\%$ in $mAP_{50}$ and $2.1\%$ in $mAP$. Similarly, removing DFM results in a drop of $1.9\%$ in $mAP_{50}$ and $2.2\%$ in $mAP$. This verifies the importance of SFM and DFM. Their fusion design, which progresses from shallow to deep levels, has exhibited a positive effect on the integration of features across various modalities.

**Effects of Different Fusion Strategies.** To verify the effectiveness of our proposed fusion strategies, we evaluate them against a standard fusion strategy Avg [81] in Table 7. Here, ("A", "B") indicates using strategy "A" and "B" to fuse high-frequency and low-frequency components, respectively. Starting with a simple baseline (Avg for fusion without DWT) and then progressively incorporating DWT along with different fusion strategies.The baseline lacks DWT and is incompatible with our improved YOLOv8 head, thus all experiments use the original head for fair comparison. By comparing the 1st and 2nd rows, we demonstrate the effectiveness of RGB-IR fusion in the wavelet domain, and further comparisons (2nd to 5th rows) show that our final combination (5th row)

| Methods | Backbone | Precision | Recall | F1 | $mAP_{50}$ | $mAP$ | Parameters | Inference time (ms) |
|---|---|---|---|---|---|---|---|---|
| (TITS'23) MFPT [85] | ResNet50 | 79.3 | 72.9 | 76.0 | 80.0 | 41.9 | 200.0M | 80.0 |
| Ours | ResNet50 | **84.2** | **77.9** | **80.9** | **86.5** | **47.9** | **193.2M** | **53.2** |
| (PRL'24) CrossFormer [22] | YOLOv5 | 78.1 | 72.8 | 75.4 | 79.3 | 42.1 | 340.0M | 80.0 |
| Ours | YOLOv5 | **83.9** | **79.7** | **81.7** | **85.8** | **47.5** | **45.6M** | **44.1** |
| YOLOv8-IR [50] | YOLOv8 | 75.1 | 65.3 | 69.9 | 72.9 | 38.3 | 43.7M | 22 |
| YOLOv8-RGB [50] | YOLOv8 | 71.6 | 62.2 | 66.6 | 66.3 | 28.2 | 43.7M | 22 |
| (GRSL'24) ESSFN [64] | YOLOv8 | 81.4 | 73.5 | 77.2 | 80.8 | 42.3 | 80.2M | 47.0 |
| Ours | YOLOv8 | **84.2** | **80.9** | **82.5** | **88.4** | **48.1** | **69.1M** | **40.0** |

Table 4. Comparison results with SOTA methods on FLIR-Aligned dataset. The best results are highlighted in **bold**.

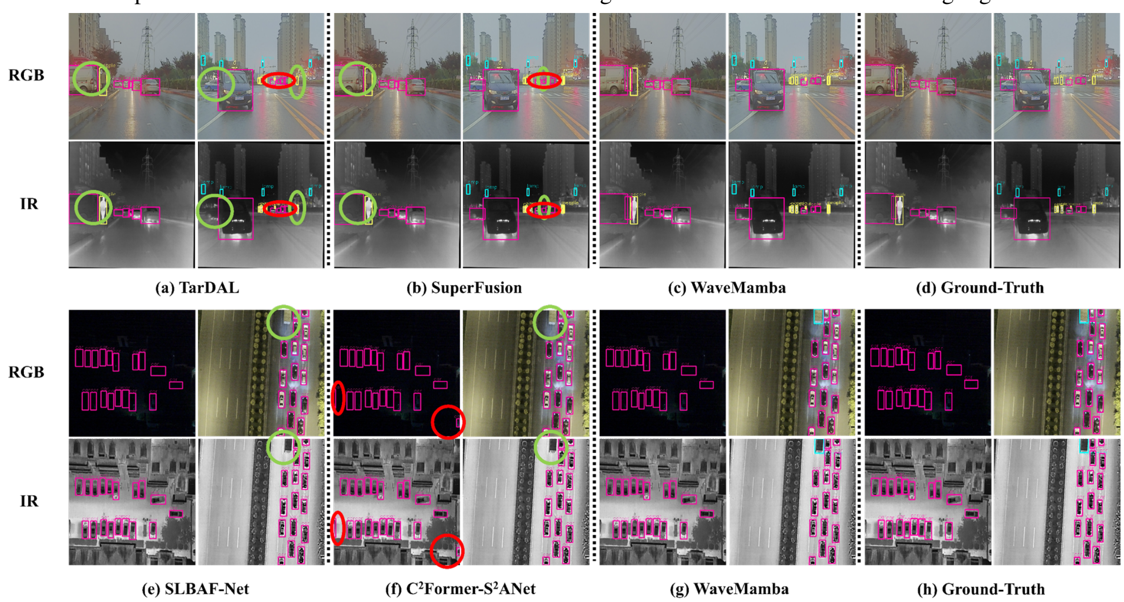


Figure 4. Detection results' visualization of several cross-modality object detection methods on $M^3$FD and DroneVehicle. Wherein, (a)-(d) present the results of $M^3$FD dataset, and (e)-(h) present the results of DroneVehicle dataset. The targets encircled by red ellipses are false positives, while those encircled by green ellipses are missed detections. Please zoom in for more details.

| Methods | $mAP_{50}$ | $mAP$ | Parameters |
|---|---|---|---|
| Origin YOLOv8 head | 91.0 | 63.3 | 76.7M |
| Improved YOLOv8 head | 92.1 | 64.4 | 69.1M |

Table 5. Effects of the improved head on $M^3$FD dataset.

| Methods | $mAP_{50}$ | $mAP$ | Parameters |
|---|---|---|---|
| w/o SFM module | 90.6 | 62.3 | 63.4M |
| w/o DFM module | 90.2 | 62.2 | 64.1M |
| ours | 92.1 | 64.4 | 69.1M |

Table 6. Effects of the SFM and DFM on $M^3$FD dataset.

achieves the best performance, effectively preserving both low-frequency spatial information and high-frequency detail characteristics of objects.

## 5. Conclusion

In this paper, we propose WaveMamba, a method that utilizes the unique and complementary frequency characteristics of different modalities to integrate features, prompting

| Methods | $mAP_{50}$ | $mAP$ | Parameters |
|---|---|---|---|
| baseline | 83.2 | 55.1 | 68.9M |
| (Avg, Avg) | 86.6 | 58.2 | 66.3M |
| (Avg, LMFB) | 90.6 | 62.6 | 76.7M |
| (HFE, Avg) | 90.4 | 62.3 | 66.3M |
| (HFE, LMFB) | 91.0 | 63.3 | 76.7M |

Table 7. Effects of different fusion strategies in WaveMamba Fusion Blocks on $M^3$FD dataset.

extensive feature interaction and improving the effectiveness of RGB-IR object detection. The main innovation of WaveMamba lies in WMFB, which incorporates an LMFB to progressively integrate low-frequency features from shallow to deep layers, alongside a HFE strategy for efficient integration of high-frequency information. Extensive experiments on four distinct types of public RGB-IR datasets demonstrate that our method achieves state-of-the-art results while balancing parameters and efficiency. As a result, this work establishes a new baseline for RGB-IR object detection, setting a precedent for future research in the field.

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